BIO392 : Seqeunce analysis introduction

Variant calling

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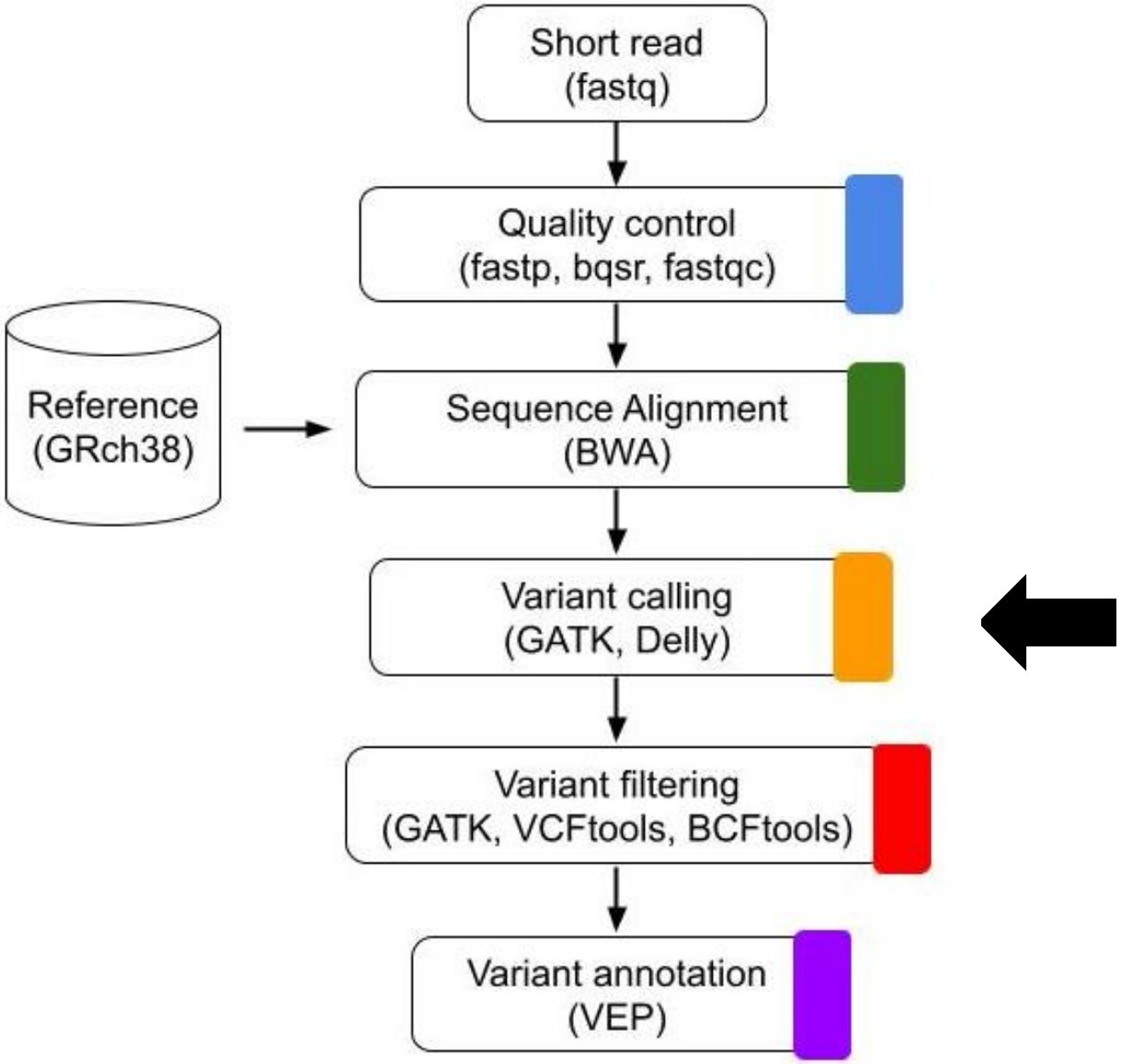
Bioinformatics workflow



What is Genetic Variants ?

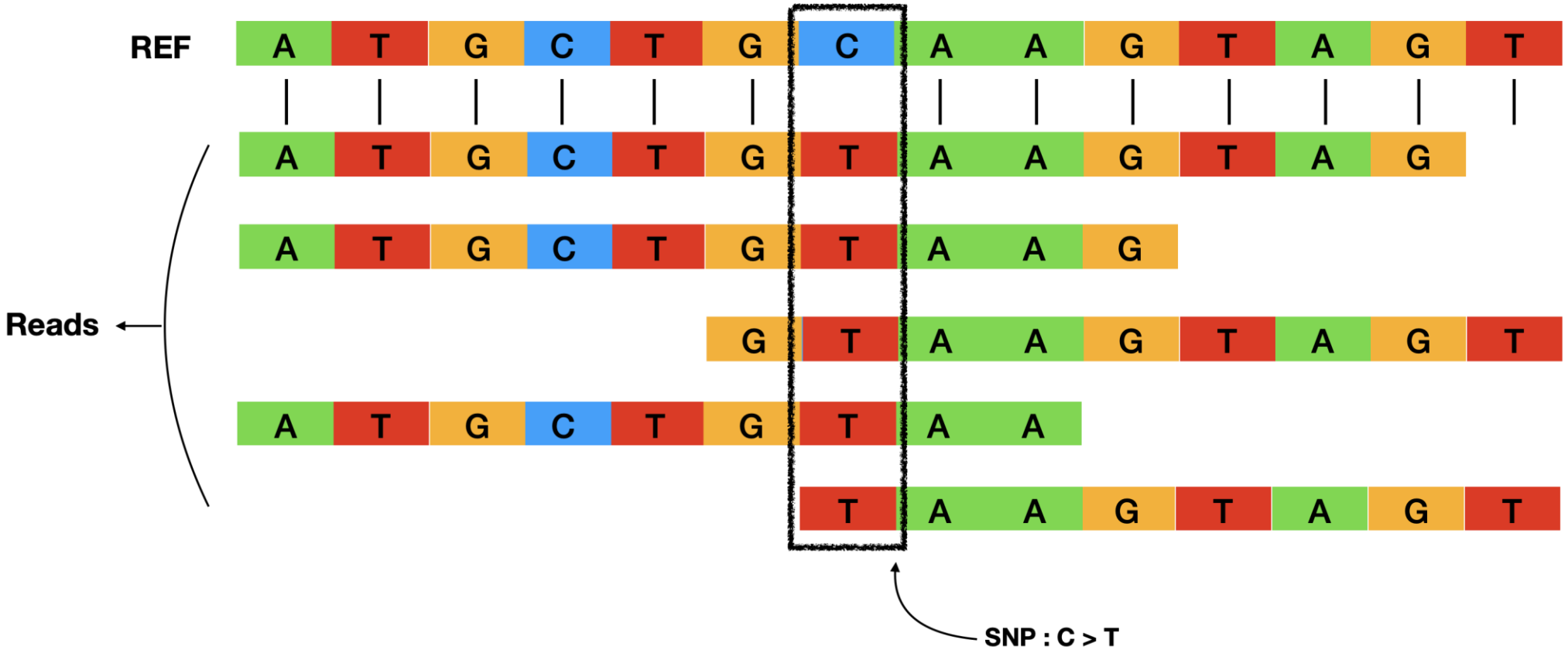
“Genetic variant are changes of nucleotide when compared to reference genome which may or may not cause positive/negative consequence”

Variants types

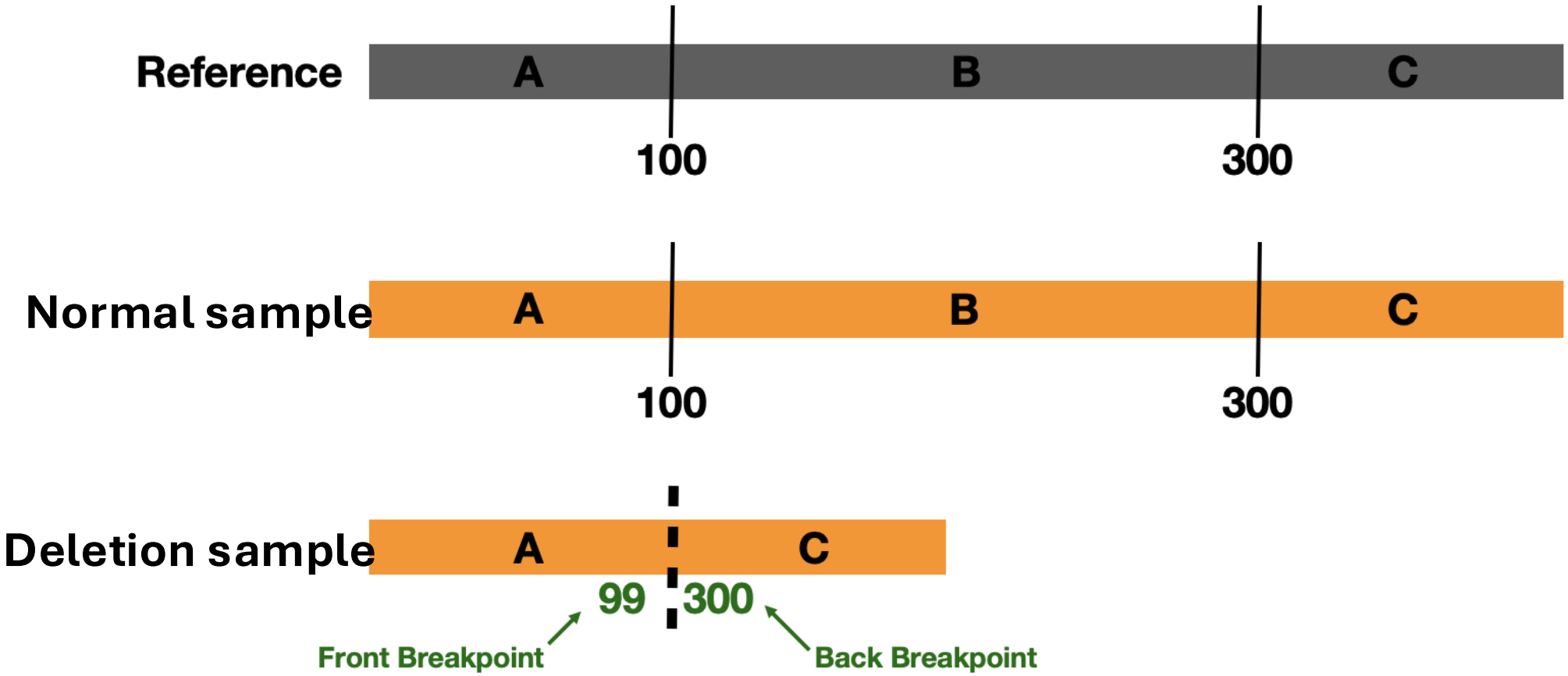
- **Small variants (<=50 bp)**
  - Single Nucleotide Polymorphism (SNP)
  - Deletion
  - Insertion
- **Structural variants (>50 bp)**
  - Deletion
  - Duplication
  - Insertion
  - Inversion

Copy number variant (CNV)

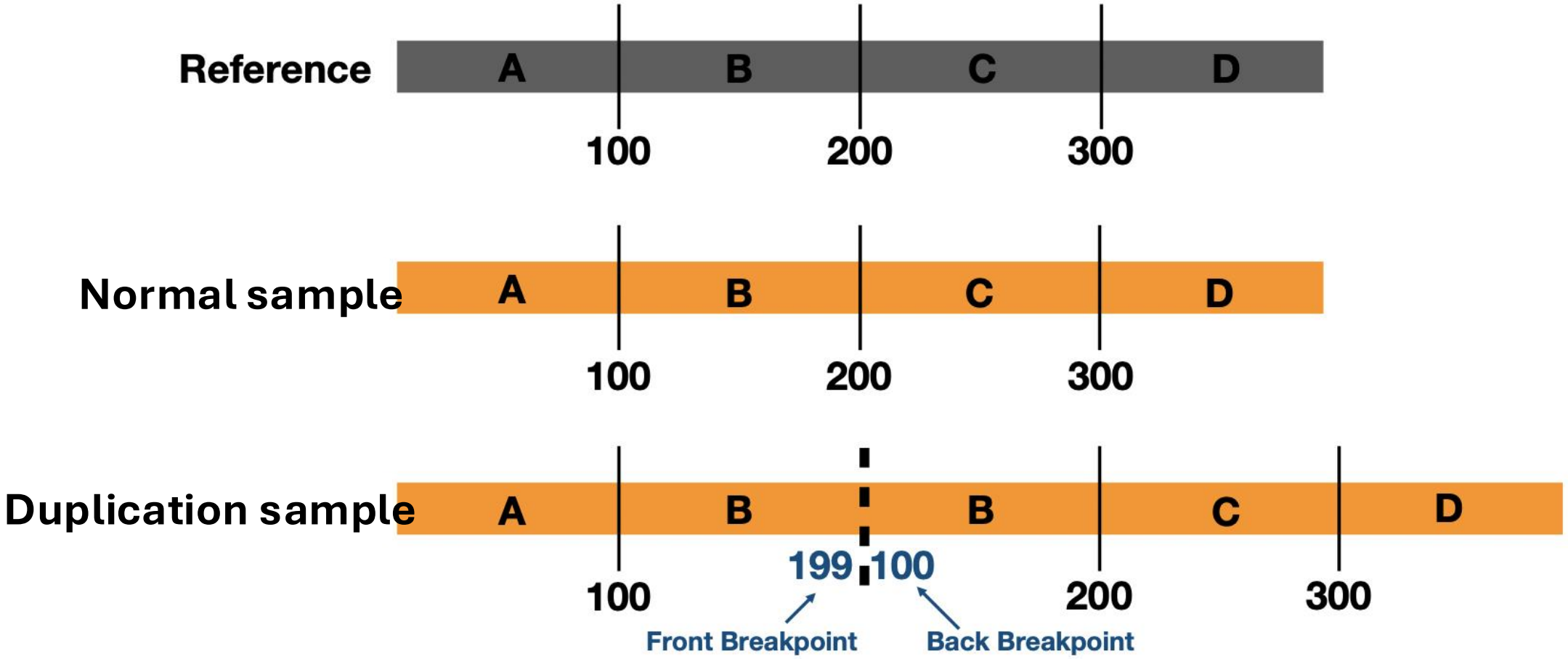
Single Nucleotide Polymorphism (SNP)



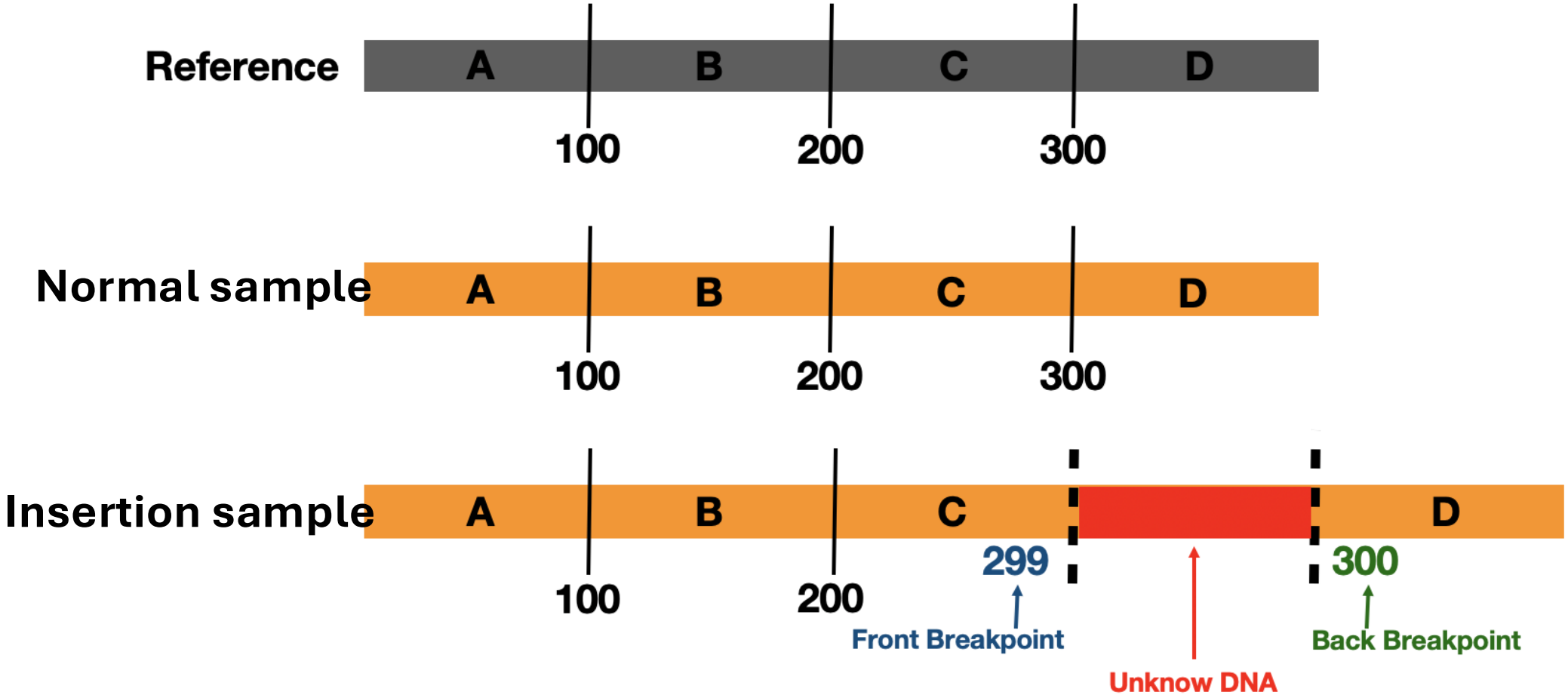
Deletion


Reference
A
B
C
100
300
Normal sample
A
B
C
100
300
Deletion sample
A
C
99
300
Front Breakpoint
Back Breakpoint

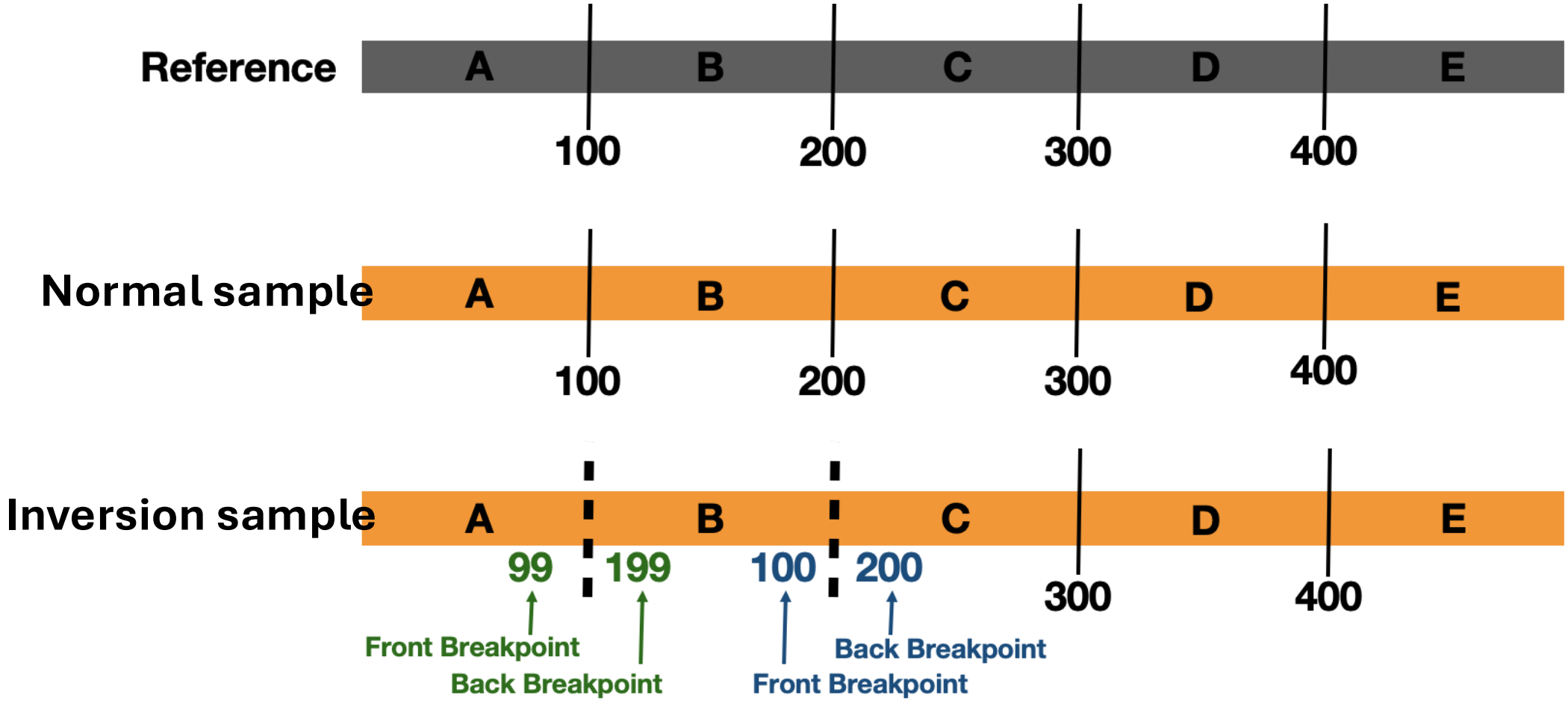
Duplication



Insertion


Reference
A
B
C
D
100
200
300
Normal sample
A
B
C
D
100
200
300
Insertion sample
A
B
C
D
100
200
299
300
Front Breakpoint
Unknow DNA
Back Breakpoint

Inversion


Reference
A
B
C
D
E
100
200
300
400
Normal sample
A
B
C
D
E
100
200
300
400
Inversion sample
A
B
C
D
E
99
199
100
200
300
400
Front Breakpoint
Back Breakpoint
Front Breakpoint
Back Breakpoint

What is VCF file?

- Human readable text file format that store variant profile.
- Use as a standard file format for most of the tools for downstream analysis.

VCF format

Header

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS      ID         REF  ALT    QUAL FILTER  INFO                              FORMAT      NA00001         NA00002         NA00003
20    14370    rs6054257  G    A      29   PASS    NS=3;DP=14;AF=0.5;DB;H2           GT:GQ:DP:HQ 0|0:48:1:51,51  1|0:48:8:51,51  1/1:43:5:.,.
20    17330    .          T    A      3    q10     NS=3;DP=11;AF=0.017               GT:GQ:DP:HQ 0|0:49:3:58,50  0|1:3:5:65,3   0/0:41:3
20    1110696  rs6040355  A    G,T    67   PASS    NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27  2|1:2:0:18,2    2/2:35:4
20    1230237  .          T    .      47   PASS    NS=3;DP=13;AA=T                   GT:GQ:DP:HQ 0|0:54:7:56,60  0|0:48:4:51,51 0/0:61:2
20    1234567  microsat1  GTC  G,GTCT 50   PASS    NS=3;DP=9;AA=G                    GT:GQ:DP    0/1:35:4        0/2:17:2        1/1:40:3
```

Chromosome

Position

Varaint ID

Reference base

Alternative allele

Quality score

Flag

Extensible Information of variation

Extensible Information of sample

Each sample information corresponding to FORMAT column

https://samtools.github.io/hts-specs/

HEADER

General information

Description of ID in INFO column

Description of ID in FORMAT column

Description of ID in FILTER column

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS     ID        REF    ALT    QUAL  FILTER   INFO   FORMAT      NA00001        NA00002        NA00003
```

Example

```
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
```

Id “DP” refer to total depth which represent as interger number

```
##FILTER=<ID=q10,Description="Quality below 10">
```

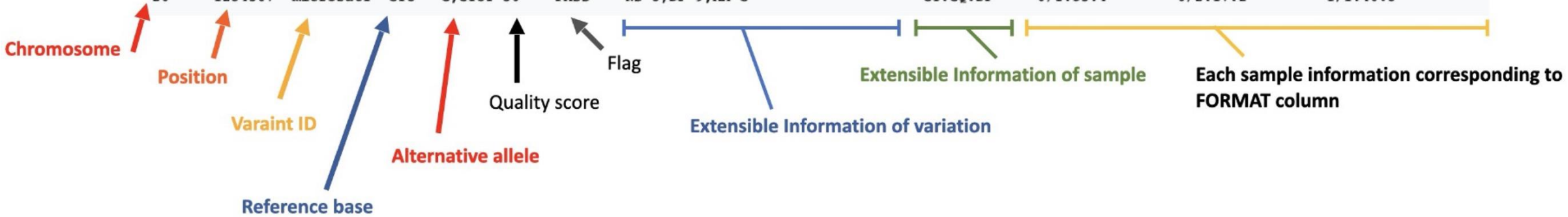
A variants that have “q10” in FILTER column is fails to pass filtering criteria due to the quality is below 10

```
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
```

Id “GT” refer to genotype which represent as string

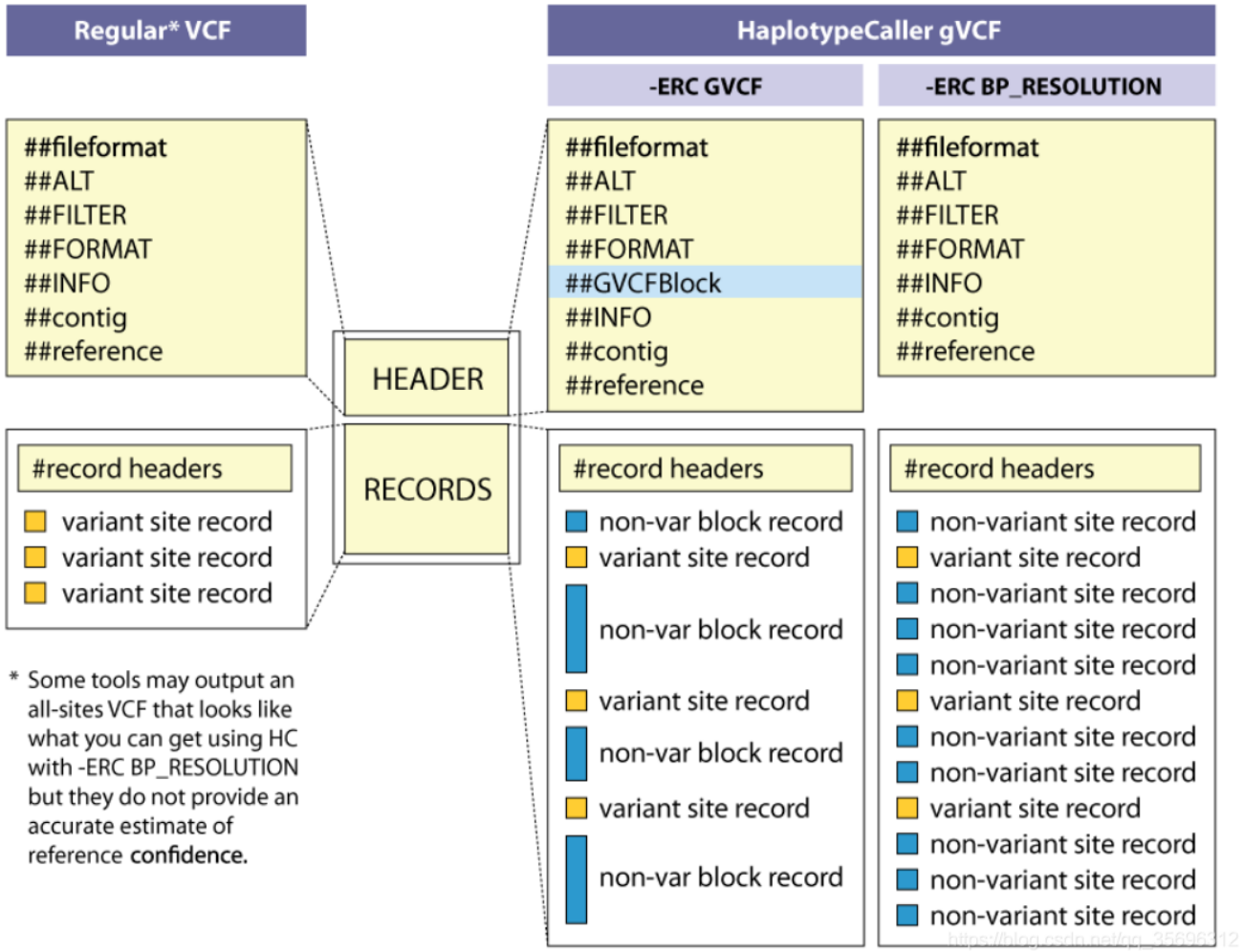
Variant records

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	NA00001	NA00002	NA00003
20	14370	rs6054257	G	A	29	PASS	NS=3;DP=14;AF=0.5;DB;H2	GT:GQ:DP:HQ	0|0:48:1:51,51	1|0:48:8:51,51	1/1:43:5:.,.
20	17330	.	T	A	3	q10	NS=3;DP=11;AF=0.017	GT:GQ:DP:HQ	0|0:49:3:58,50	0|1:3:5:65,3	0/0:41:3
20	1110696	rs6040355	A	G,T	67	PASS	NS=2;DP=10;AF=0.333,0.667;AA=T;DB	GT:GQ:DP:HQ	1|2:21:6:23,27	2|1:2:0:18,2	2/2:35:4
20	1230237	.	T	.	47	PASS	NS=3;DP=13;AA=T	GT:GQ:DP:HQ	0|0:54:7:56,60	0|0:48:4:51,51	0/0:61:2
20	1234567	microsat1	GTC	G,GTCT	50	PASS	NS=3;DP=9;AA=G	GT:GQ:DP	0/1:35:4	0/2:17:2	1/1:40:3



1. **CHROM**: The chromosome (contig) on which the variant occurs
2. **POS**: The genomic coordinates on which the variant occurs. For deletions, the position given here are on of the bases preceding the event.
3. **ID:** An identifier for the variant (if it exists). Typically a dbSNP database if that is known.
4. **REF**: The reference allele on the forward strand.
5. **ALT:** The alternate allele(s) on the forward strand. More than one may be present.
6. **QUAL:** A probability that the REF/ALT variant exists at this site. It is in Phred scale, just as the FASTQ quality and the MAPQ field in the SAM file are.
7. **FILTER:** The name of filters that the variant fails to pass, or the value PASS if the variant passed all filters. If the FILTER value is ., then no filtering has been applied to the record.
8. **INFO:** Contains the site-specific annotations represented as ID=VALUE format.
9. **FORMAT:** Sample-level annotations as colon separated TAGS.

What is GVCF?


Regular* VCF
HaplotypeCaller gVCF
-ERC GVCF
-ERC BP_RESOLUTION
##fileformat
##ALT
##FILTER
##FORMAT
##INFO
##contig
##reference
##GVCFBlock
HEADER
RECORDS
#record headers
variant site record
non-var block record
non-variant site record
* Some tools may output an all-sites VCF that looks like what you can get using HC with -ERC BP_RESOLUTION but they do not provide an accurate estimate of reference confidence.

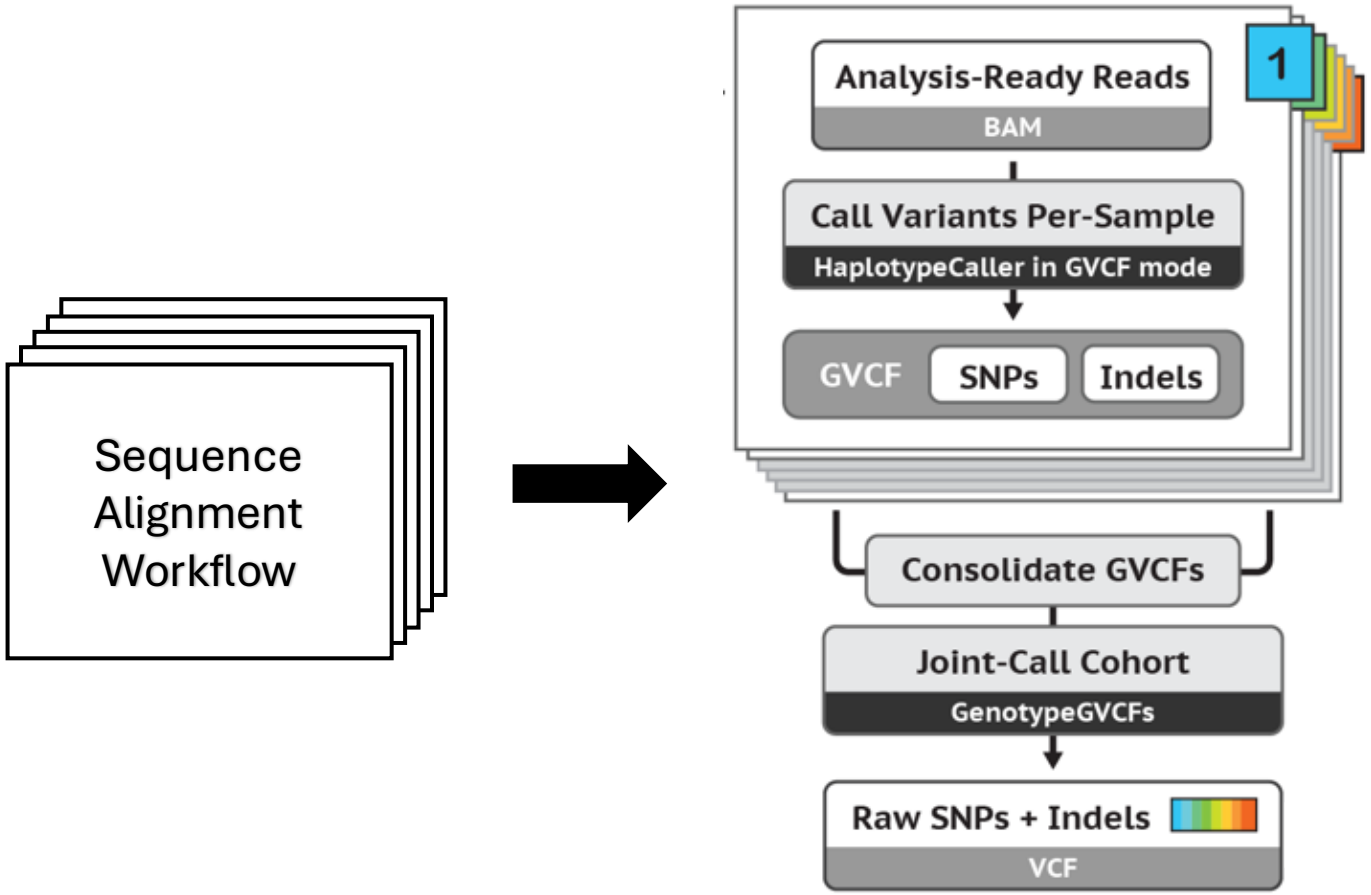
Small variant calling

- Input
  - Sequence alingment file (SAM/BAM)
  - Reference file (Fasta file)

- Variants caller
  - GATK4 (https://gatk.broadinstitute.org/hc/en-us)

- Output
  - VCF file

Genome Analysis Tools Kit (GATK)



Command

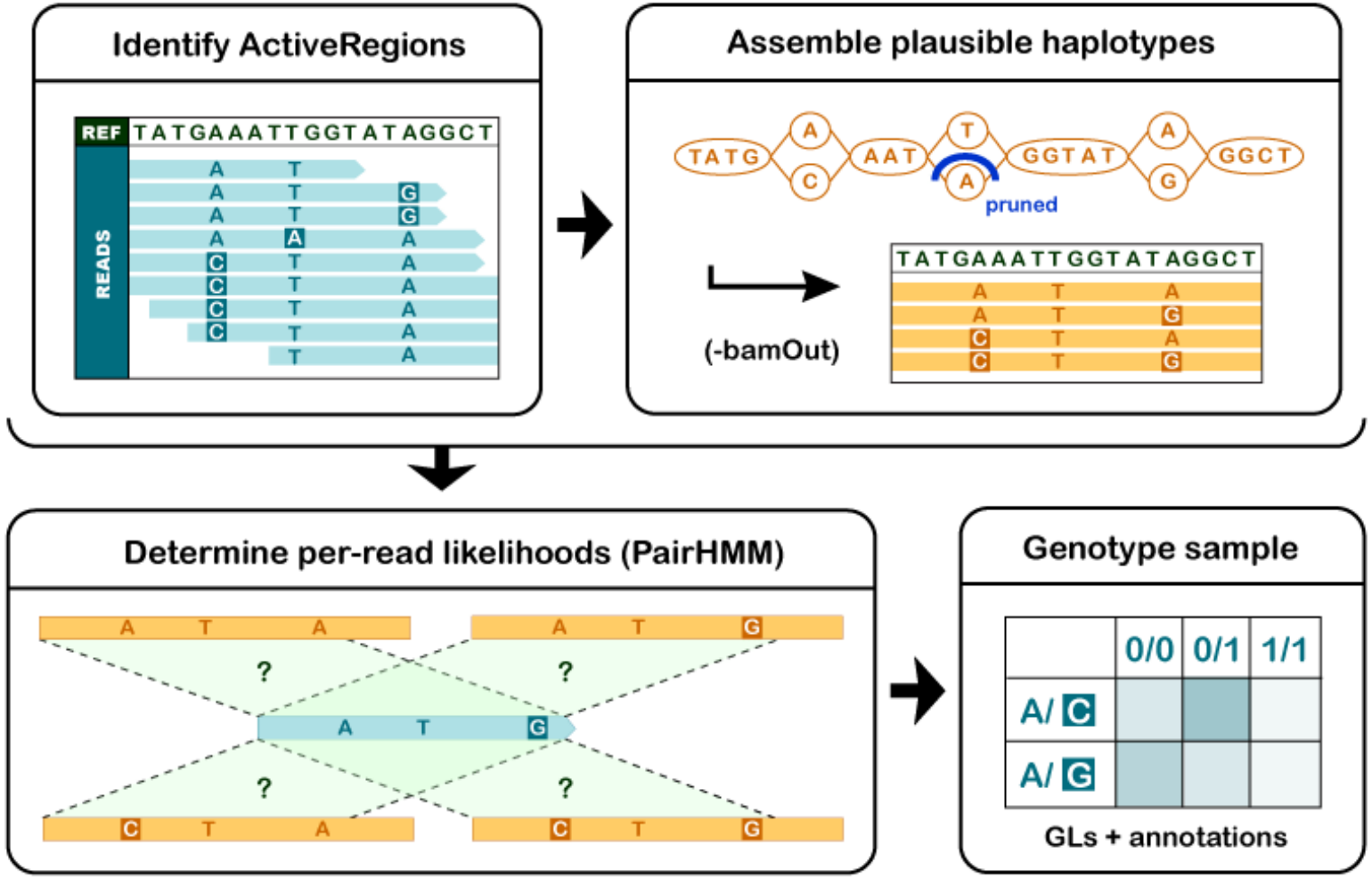
```
gatk [SomeToolsName] [options]
```

Tools

- HaplotypeCaller
- GenomicsDBImport
- GenotypeGVCFs

GATK HaplotypeCaller Algorithm

Call SNPs and indels via
local re-assembly of haplotypes



GATK HaplotypeCaller

```
gatk --java-options "-Xmx${task.memory.toGiga()}G" HaplotypeCaller \
      -R ${reference file} \
      -I $bam \
      -ERC GVCF \
      -O ${sampleId}.g.vcf \
      --tmp-dir .
```

-R : Refernece sequence file.

--java-options “-Xmx16G”

-I : BAM/SAM/CRAM/ file containing reads.

-O : File to which variants should be written.

-ERC : Mode for emitting reference confidence scores. (output will contain non-variant region)

--tmp-dir : Temp directory to use.

GATK GenomicDBImport (multiple sample)

Use to merge GVCFs from multiple samples into GenomicsDB before joint genotyping.

```
gatk --java-options "-Xmx${task.memory.toGiga()}G -Xms${task.memory.toGiga()}G" GenomicsDBImport \
      --genomicsdb-workspace-path genomic_database \
      --batch-size 50 \
      -L Chromosome \
      --sample-name-map samples.mapping \
      --tmp-dir=tmp \
      --reader-threads ${task.cpus}
```

--genomicsdb-workspace-path : Workspace for GenomicsDB. (Path where GenomicDB will take place. Must be an empty or non-existent directory)
--L : One or more genomic intervals over which to operate.
--sample-name-map : Path to tab delimited file containing sample name and path to sample vcf.
--batch-size : Number of sample which reader are open at once. (0 is mean all sample at once)
--reader-threads : How many simultaneous threads to use when opening VCFs in batches.
--tmp-dir : Temp directory to use.

samples.mapping

sample1	sample1.vcf.gz
sample2	sample2.vcf.gz
sample3	sample3.vcf.gz

GATK GenotypeGVCFS

Perform joint genotyping on samples pre-called with HaplotypeCaller.
the input samples must possess genotype likelihoods produced by HaplotypeCaller with `-ERC GVCF` or `-ERC BP_RESOLUTION

Single sample GVCF

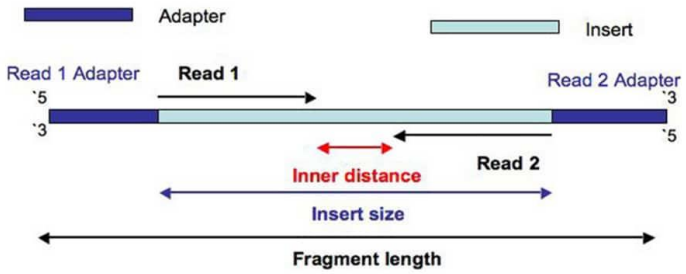
```
gatk --java-options "-Xmx${task.memory.toGiga()}G" GenotypeGVCFs \
    -R ${params.reference} \
    -V $gvcf \
    -O ${sampleId}.vcf.gz
```

-R : Reference sequence file
-V : GVCF file or path to GenomicDB workspace
-0 : File to which variants should be written
--tmp-dir : Temp directory to use.

Multiple sample

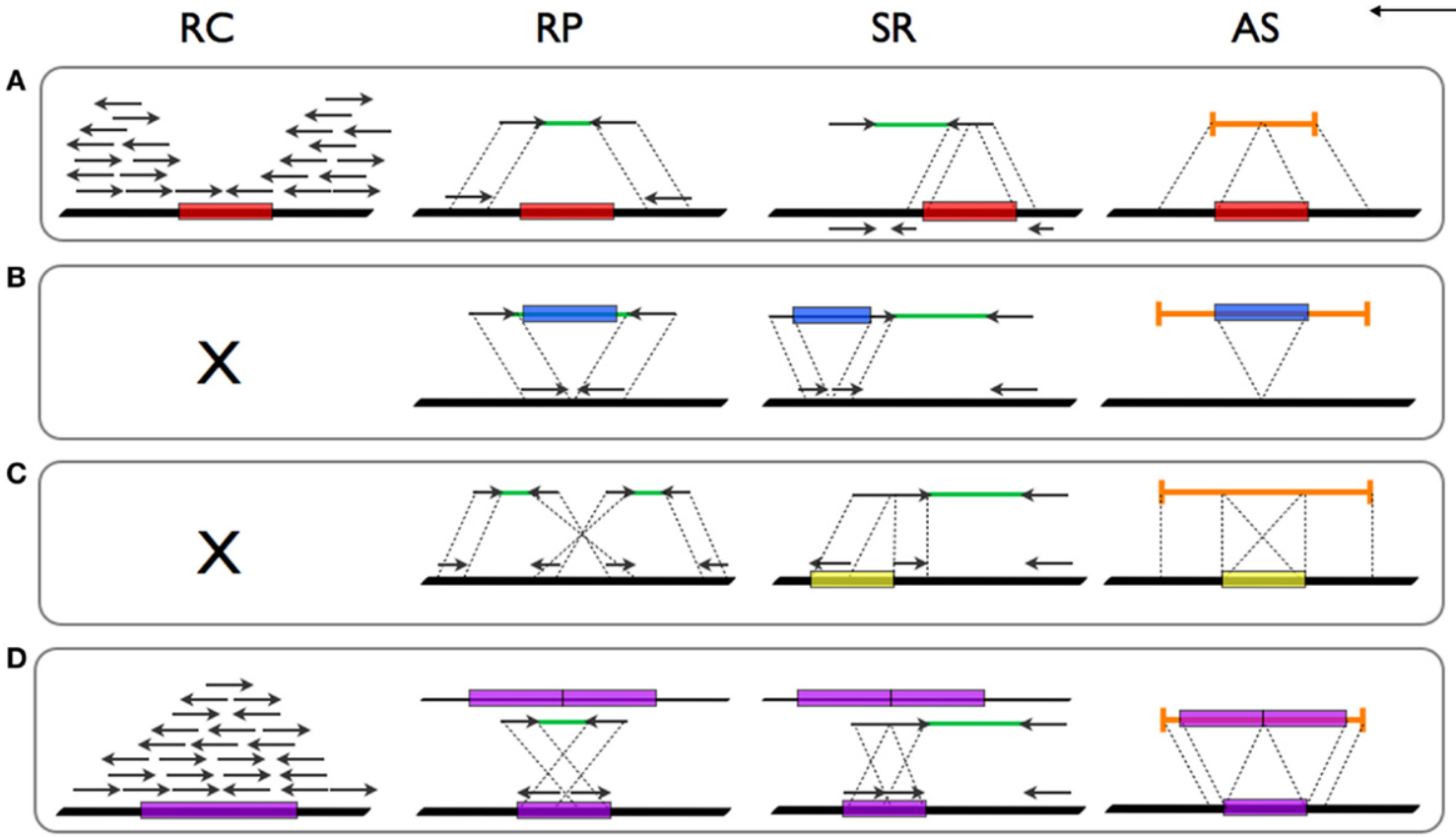
```
gatk --java-options "-Xmx${task.memory.toGiga()}G -Xms${task.memory.toGiga()}G" GenotypeGVCFs \
    -R ${params.reference} \
    -V gendb://${databaseDir} \
    -O combined.vcf \
    --tmp-dir=tmp
```

Structural Variant Calling



4 stratergies

RC = Read count
RP = Read pair
SR = Split read
AS = Assembly

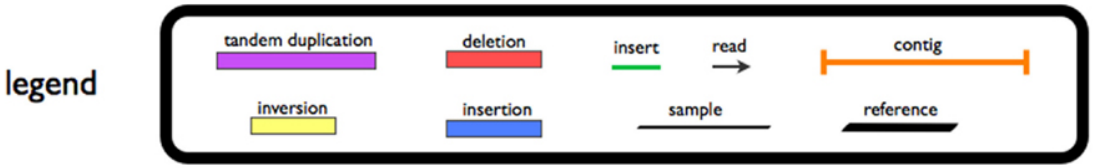


Deletion

Insertion

Inversion

Duplication



Tattini L, D'Aurizio R, Magi A. Detection of Genomic Structural Variants from Next-Generation Sequencing Data.

Delly

https://github.com/dellytools/delly

- Structural variant caller using paired-ends , split-reads strategies and read-depth strategies
- Detected variant classes
  - Copy number variant (CNV)
  - Deletions
  - Insertions
    - Fully-assembled insertions
    - Partially-assembled (ie. inferred) insertions
  - Inversions
  - Tandem Duplications
  - Inter-chromosomal Translocations

Delly

Structural variant calling

```
delly call -o ${sampleId}_dellySV.bcf \
    -g ${params.reference} \
    ${bam}
```

--g : Refernece Fasta file.
--o : Output BCF format

Copy number variant calling

```
delly cnv -o ${sampleId}_dellyCNV.bcf \
    -g ${params.reference} \
    -l ${sampleId}_dellySV.bcf \
    -m ${params.delly_reference} \
    -c ${sampleId}_cov.gz \
    ${bam}
```

--g : Refernece Fasta file.
--o : Output BCF format.
--l : Input delly SV calls.
--c : Output coverage.
--m : Mappability map file (https://gear-genomics.embl.de/data/delly/)

Delly

- Delly also have some unique INFO and FORMAT field.

- INFO filed
  - **IMPRECISE** : Flag indicating that the structural variation is imprecise, i.e. the exact breakpoint location is not found
  - **PRECISE** : Flag indicating that the structural variation have precise breakpoint.
  - **PR** : Pair-end support structural variant.
  - **SR** : Split read support structural variant.
- FORMAT fields
  - **CN :** Integer copy-number.
  - **CNL :** Log10-scaled copy-number likelihood.
  - **RR** : Reads support reference junction.
  - **RV** : Reads support structural variant junction.
  - **FT** : Sample filter, 'PASS' indicates that all filters have passed for this sample

Post-processing

- Filter out low confident variants
- Filter out variants for specific purpose
- Manipulate variants information

- Tools
  - GATK
  - VCFtools (http://vcftools.sourceforge.net)

Variant Quality Score Recalibration (VQSR)

- Use known highly validate variant information and annotation profile to train machine learning model.
- The model will learn the rule to recognize how good and bad variant look like.
- The threshold was pick based on the truth sensitivity.

Variant Quality Score Recalibration (VQSR)

Frist, Create model with VariantRecalibrator

```
gatk --java-options "-Xmx8G" VariantRecalibrator \
    -R ${Reference_file} \
    -V ${input_vcf} \
    --resource:${resource_name},known=${true/false},training=${true/false},truth=${true/false},prior=${prior_value} ${known_variant_file}  \
    -an QD -an MQ -an MQRankSum -an ReadPosRankSum -an FS -an SOR  \
    -mode ${SNP/INDEL} \
    -O ${output_recal_file} \
    --tranches-file ${out_put_tranches}
```

-R: reference file.
-V: input VCF file.
-an: annotation value locate in INFO column. (Suggest QD, MQ, MQRankSum, ReadPosRankSum, FS and SOR)
-mode: Recalibration mode. (SNP, INDEL, BOTH)
-O : output recalibration file.
--tranches-file : output tranches file.
--resource : known variant site. (can be VCF file)

Variant Quality Score Recalibration (VQSR)

Resource dataset

Truth resource : A call set that has been validate to very high degree of confidence. Consider as true sites (truth=true). The site can be use to train (train=true) and choose threshold for filtering variant.

Training resource : A call set that has been validate in some degree of confidence. May contain false positive and true variants (truth=false). Use to train the recalibration model (training=true).

Known sites resource : A call set that has not been validate (truth=false). The program will not use this resource to train model (train=false). Program will use these to stratify output metrics such as Ti/Tv ratio by whether variants are present in dbsnp or not. (known=true)

https://console.cloud.google.com/storage/browser/gcp-public-data--broad-references

Variant Quality Score Recalibration (VQSR)

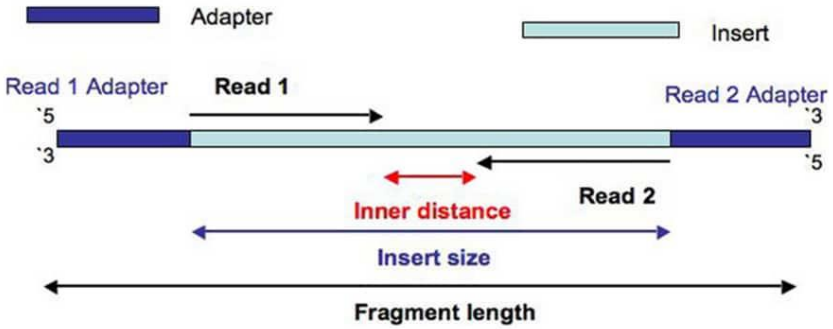
Annotation profile : QualByDepth (QD)

This is the variant confidence (from the QUAL field) divided by the unfiltered depth of non-homozygous reference samples. This annotation is intended to normalize the variant quality in order to avoid inflation caused when there is deep coverage. For filtering purposes it is better to use QD than either QUAL or DP directly. (QD = QUAL/DP)

Variant Quality Score Recalibration (VQSR)

Annotation profile : FisherStrand (FS)

This is the Phred-scaled probability that there is strand bias at the site. Strand Bias tells us whether the alternate allele was seen more or less often on the forward or reverse strand than the reference allele. When there little to no strand bias at the site, the FS value will be close to 0.



Variant Quality Score Recalibration (VQSR)

Annotation profile : StrandOddRatio (SOR)

This is another way to estimate strand bias using a test similar to the symmetric odds ratio test. SOR was created because FS tends to penalize variants that occur at the ends of exons. Reads at the ends of exons tend to only be covered by reads in one direction and FS gives those variants a bad score. SOR will take into account the ratios of reads that cover both alleles. (lower is better) (good to check is your data are WES or targeted sequencing)

Variant Quality Score Recalibration (VQSR)

Annotation profile : RMSMappingQuality (MQ)

This is a root mean square mapping quality over all the read at variant site. Better than consider average mapping quality. When the mapping qualities are good at a site, the MQ will be around 60.

Annotation profile : MappingQualityRankSumTest (MQRankSum)

This is Rank Sum Test for mapping qualities of REF versus ALT reads. It compare mapping quality of read support reference allele and alternate allele. A negative value indicates that the reads supporting the alternate allele have lower mapping quality scores than those supporting the reference allele and vice versa. The ideal value is zero.

Variant Quality Score Recalibration (VQSR)

Annotation profile : ReadPosRankSumTest (ReadPosRankSum)

This is Rank Sum Test for site position within reads. It compares whether the positions of the reference and alternate alleles are different within the reads. Seeing an allele only near the ends of reads is indicative of error, because that is where sequencers tend to make the most errors. A negative value indicates that the alternate allele is found at the ends of reads more often than the reference allele and vice versa. The ideal value is zero.

Variant Quality Score Recalibration (VQSR)

Second, Apply model with ApplyVQSR

```
gatk --java-options "-Xmx8G" ApplyVQSR \
  -V ${input_vcf} \
  -mode ${SNP/INDEL} \
  --recal-file ${recal_file} \
  --tranches-file ${tranches_file} \
  --truth-sensitivity-filter-level ${percent_sensitivity} \
  --create-output-variant-index true \
  -O ${output_vcf}
```

-V: input VCF file.
-mode: Recalibration mode. (SNP, INDEL, BOTH)
-O : output VCF file.
--recal-file : recal file.
--truth-sensitivity-filter-level : truth sensitivity level.
--tranches-file : tranches file.
--create-output-variant-index : create index for output VCF. (true/false)

GATK Hard-filtering command (Alternative)

```
gatk VariantFiltration \
     -V input_vcf \
     -O output_vcf \
     -filter "QD < 2.0" --filter-name "QD2" \
     -filter "MQ < 40.0" --filter-name "MQ40" \
     -filter "FS > 60.0" --filter-name "FS60" \
     -filter "SOR > 3.0" --filter-name "SOR3" \
     -filter "MQRankSum < -12.5" --filter-name "MQRankSum-12.5" \
     -filter "ReadPosRankSum < -8.0" --filter-name "ReadPosRankSum-8"
```

-V : Input VCF file
-O : Output VCF file

-filter : Perform Filtering on specific INFO field value with specific criteria.
--filter-name : Define this parameter after "-filter" will tell program to add value that has been define for this parameter on Filter column in VCF file if variant fail criteria. In the other hand "PASS" will be added.

https://gatk.broadinstitute.org/hc/en-us/articles/360035890471-Hard-filtering-germline-short-variants

Filter with VCFtools

https://vcftools.sourceforge.net/

This toolset can be used to perform the following operations on VCF files:

- Filter out specific variants
- Compare VCF files
- Summarize variants
- Convert to different file types

VCFtools

vcftools [**--vcf** FILE | **--gzvcf** FILE | **--bcf** FILE] [**--out** OUTPUT PREFIX] [FILTERING OPTIONS] [OUTPUT OPTIONS]

```
vcftools --gzvcf inputGVCF \
    --min-alleles 2 \
    --max-alleles 2 \
    --recode \
    --recode-INFO-all \
    --remove-filtered-all \
    --out output_prefix
```

--recode
--recode-bcf
These options are used to generate a new file in either VCF or BCF from the input VCF or BCF file after applying the filtering options specified by the user. The output file has the suffix ".recode.vcf" or ".recode.bcf". By default, the INFO fields are removed from the output file.
--recode-INFO *<string>*
--recode-INFO-all
These options can be used with the above recode options to define an INFO key name to keep in the output file. This option can be used multiple times to keep more of the INFO fields. The second option is used to keep all INFO values in the original file.

--remove-filtered-all
Removes all sites with a FILTER flag other than PASS.

--min-alleles *<integer>*
--max-alleles *<integer>*
Include only sites with a number of alleles greater than or equal to the "--min-alleles" value and less than or equal to the "--max-alleles" value. One of these options may be used without the other. For example, to include only bi-allelic sites, one could use:

BCFtools

https://samtools.github.io/bcftools/bcftools.html

A utilities tools use to manipulate variant call in Variant Call Format (VCF) and its binary (BCF).

- Edit VCF/BCF files.
- Filter VCF/BCF files.
- Convert VCF/BCF to other formats.
- Concatenate VCF/BCF files.
- Merge VCF/BCF files.

LIST OF COMMANDS

For a full list of available commands, run **bcftools** without arguments. For a full list of a

- annotate .. edit VCF files, add or remove annotations
- call .. SNP/indel calling (former "view")
- cnv .. Copy Number Variation caller
- concat .. concatenate VCF/BCF files from the same set of samples
- consensus .. create consensus sequence by applying VCF variants
- convert .. convert VCF/BCF to other formats and back
- csq .. haplotype aware consequence caller
- filter .. filter VCF/BCF files using fixed thresholds
- gtcheck .. check sample concordance, detect sample swaps and contamination
- head .. view VCF/BCF file headers
- index .. index VCF/BCF
- isec .. intersections of VCF/BCF files
- merge .. merge VCF/BCF files files from non-overlapping sample sets
- mpileup .. multi-way pileup producing genotype likelihoods
- norm .. normalize indels
- plugin .. run user-defined plugin
- polysomy .. detect contaminations and whole-chromosome aberrations
- query .. transform VCF/BCF into user-defined formats
- reheader .. modify VCF/BCF header, change sample names
- roh .. identify runs of homo/auto-zygosity
- sort .. sort VCF/BCF files
- stats .. produce VCF/BCF stats (former vcfcheck)
- view .. subset, filter and convert VCF and BCF files

https://samtools.github.io/bcftools/bcftools.html#expressions

BCFtools

```
bcftools view -f PASS \
    -e 'INFO/IMPRECISE=1' \
    -o output_file \
    input_file
```

Command : View
Use for visulize file, filter and conver VCF and BCF files

-f : Skip sites where FILTER column does not contain any of the strings specify.

-e : Exclude sites for which the expression is true.

-o : Output file name.

Q&A